

# AquaPress AP-210 Pressure Transmitter -- Technical Manual

## PRODUCT OVERVIEW

The AquaPress AP-210 is a state-of-the-art pressure transmitter designed specifically for monitoring discharge pressure in pump stations within water distribution networks. Engineered for reliability and precision, the AquaPress AP-210 utilizes advanced sensor technology to deliver consistent and accurate pressure readings, ensuring optimal performance of pumping systems. Its robust construction and simple installation make it an ideal choice for a variety of applications in water and wastewater treatment facilities.

## TECHNICAL SPECIFICATIONS

- Measurement Range: 0 to 500 psi (0 to 34.5 bar)
- Accuracy:  $\pm 0.5\%$  of Full Scale (FS) at 25°C
- Output Signal: 4-20 mA (two-wire) or RS-485 digital communication (Modbus RTU)
- Process Connection: 1/2" NPT (male) or 3/4" BSP (male)
- Ingress Protection: IP68, suitable for submersible applications up to 10 meters

The AquaPress AP-210 is designed to operate in harsh environments, ensuring durability and long-lasting performance in demanding conditions. Its wide measurement range and high accuracy make it suitable for a variety of pressure monitoring tasks.

## CALIBRATION PROCEDURE

The AquaPress AP-210 requires regular calibration to maintain its accuracy. The calibration procedure involves zero and span adjustments, which should be performed using a calibrated pressure source and a digital multimeter.

### 1. Zero Adjustment:

- Disconnect the AquaPress AP-210 from the process line.
- Apply a pressure of 0 psi (0 bar) using a calibrated pressure source.
- Connect a multimeter to monitor the output signal.
- Adjust the zero setting on the transmitter until the output signal reads 4 mA.
- Verify the adjustment by rechecking at 0 psi.

### 2. Span Adjustment:

- After completing the zero adjustment, apply a known pressure (e.g., 500 psi / 34.5 bar).
- Monitor the output signal with the multimeter.
- Adjust the span setting on the transmitter until the output signal reads 20 mA.
- Remove the pressure source and ensure the output returns to 4 mA at 0 psi.
- Document the calibration readings for future reference.

## ERROR CODES

The AquaPress AP-210 is equipped with an error code diagnostic feature to enhance troubleshooting efficiency. Below are common error codes, their causes, and recommended resolutions:

- Error Code E1: Sensor Fault

Cause: Internal sensor malfunction.

Resolution: Check wiring connections. If the issue persists, replace the unit.

- Error Code E2: Over Pressure

Cause: Input pressure exceeds maximum rated value.

Resolution: Reduce process pressure and inspect for leaks. Verify system pressure.

- Error Code E3: Under Pressure

Cause: Input pressure is below the minimum required threshold.

Resolution: Confirm that there is no blockage in the process line. Verify the pressure source.

- Error Code E4: Communication Error

Cause: Failure in digital communication (for RS-485 models).

Resolution: Check wiring and connections. Confirm Modbus settings are correct.

- Error Code E5: Calibration Error

Cause: Calibration values are out of specification.

Resolution: Perform recalibration following the calibration procedure outlined above.

## MAINTENANCE

To ensure the longevity and performance of the AquaPress AP-210, regular maintenance is essential. Follow these maintenance guidelines:

1. Visual Inspection: Perform a routine visual inspection of the transmitter for signs of damage, corrosion, or leaks. Ensure that all connections are secure.
2. Clean the Process Connection: Regularly clean the process connection to prevent blockages from debris or buildup, which could affect accuracy. Use appropriate cleaning solutions compatible with the application.
3. Functionality Test: Conduct a functionality test at least once every six months to verify the accuracy of the readings. Compare the output against a calibrated reference pressure gauge.
4. Documentation: Maintain a log of all maintenance activities, including calibration dates, error codes encountered, and corrective actions taken. This documentation aids in identifying trends that may require attention.
5. Replacement Schedule: Replace the AquaPress AP-210 every five years, or sooner if any performance issues are noted, to ensure optimal operation and reliability.

By adhering to these maintenance procedures, operators can maximize the operational lifespan and reliability of the AquaPress AP-210, thereby enhancing the efficiency of their water distribution networks.